

Magnets

magnets are bipolar

they have two poles; north & south

like poles repel

opposite/unlike poles attract

* magnetised materials can attract & repel and are always attracted to the magnet regardless of which pole is held close to it

* unmagnetised materials can be attracted to a magnet but will never experience repulsion

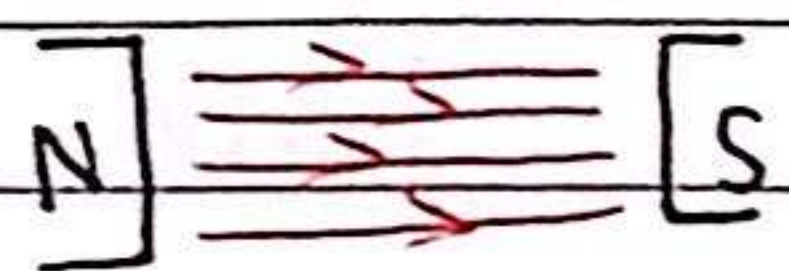
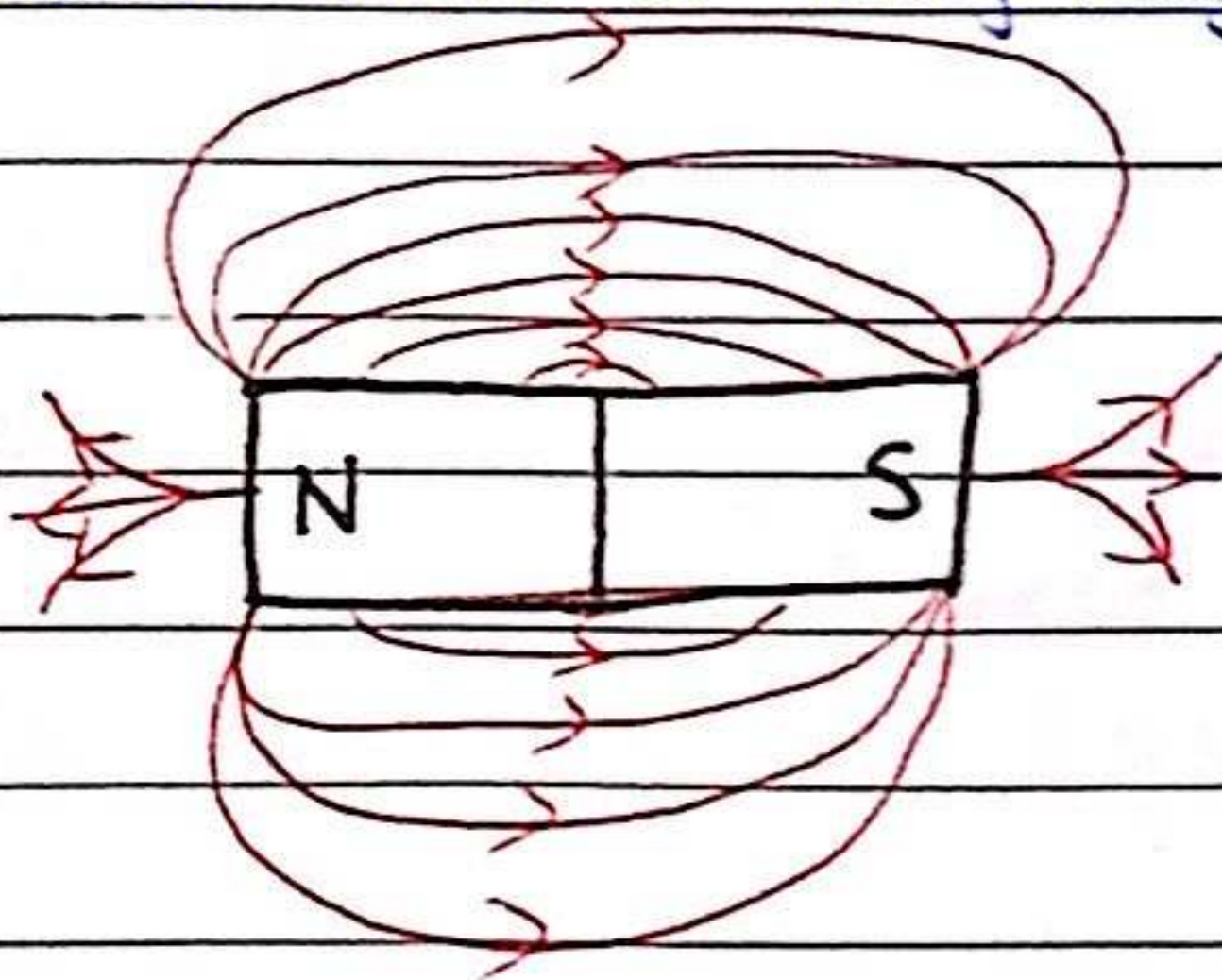
Magnetic field

a region where magnetic material materials experience force

drawn on by using lines from North to South

space between lines represents field strength

→ induced magnets are unmagnetised objects that gain temporary magnetic capabilities when it is put inside the magnetic field of any magnet.



* magnetic materials are attracted to magnets while non-magnetic materials are not

permanent magnets (steel)

produce their own magnetic field

don't lose their magnetism

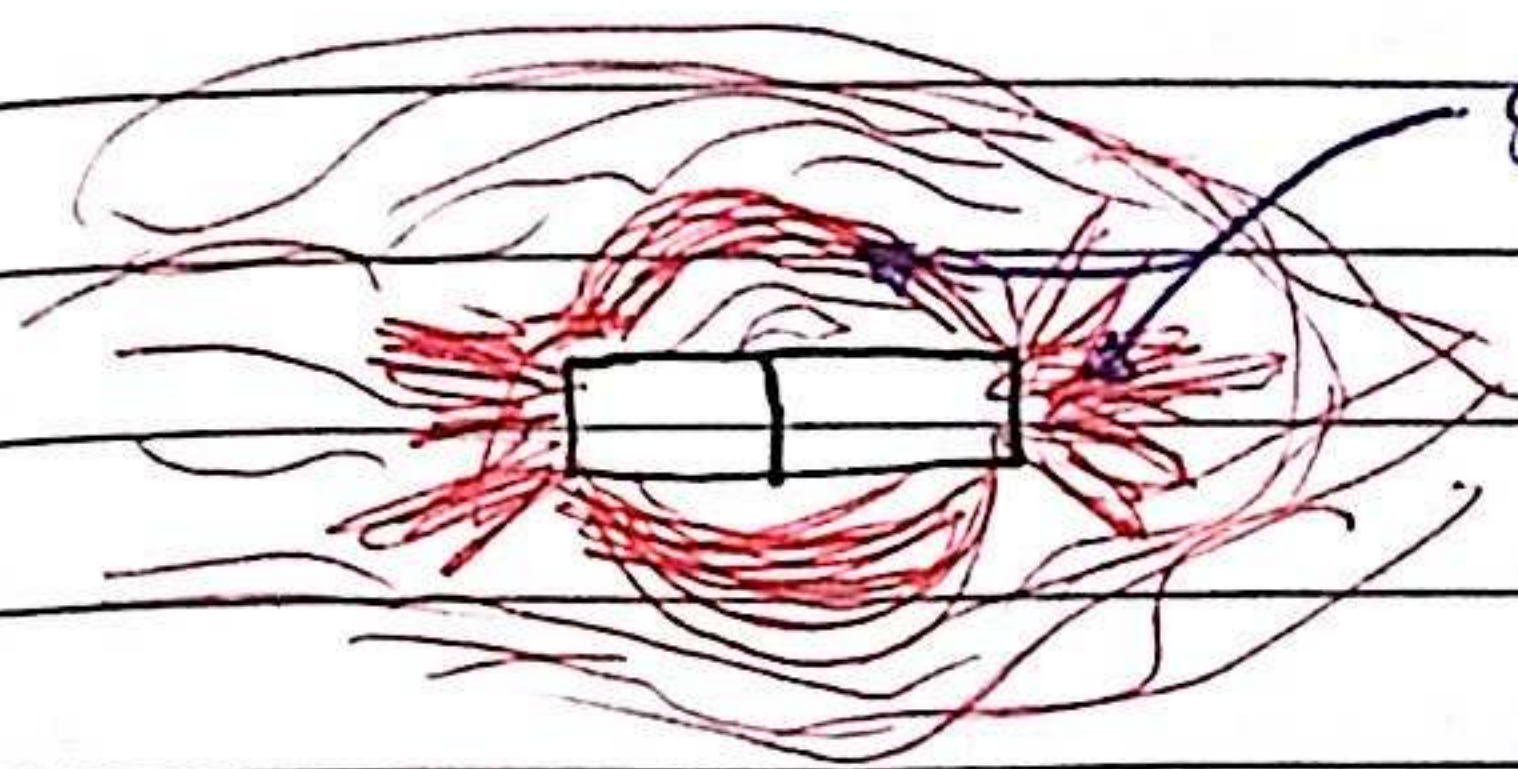
temporary magnets (soft iron)

when placed in a magnetic field, they turn into a magnet temporarily

always attracted to a permanent magnet

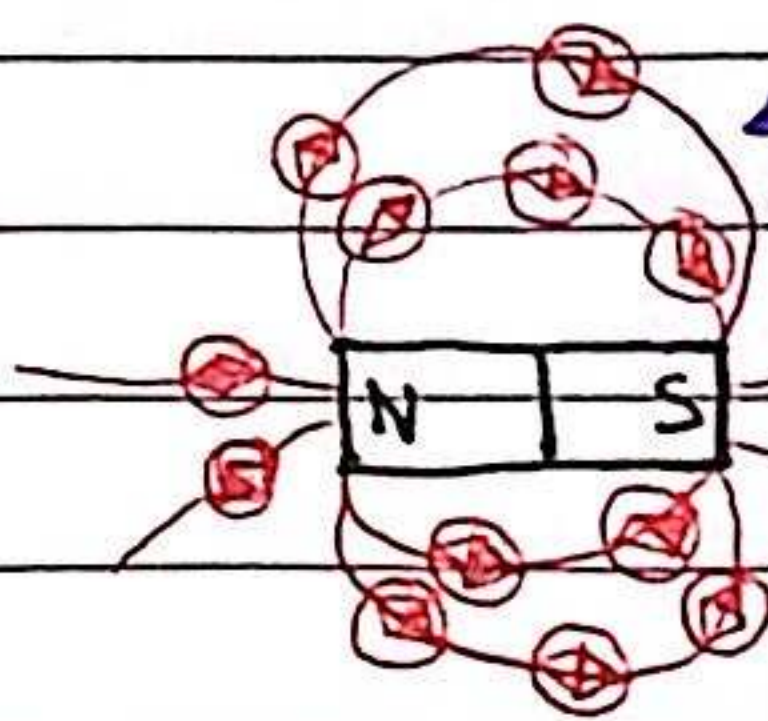
Plotting magnetic field lines

iron filings:-



Strong magnetic field
(doesn't show 'N' & 'S')

Compasses:-



Compasses showing the
direction of the magnetic
field

(doesn't show the strength)

permanent magnets

usually (but not always) made from steel

tend to stay magnetised

e.g. compasses, toys, fridge magnets etc.

Electromagnets

use electricity to create a magnet from a current-carrying wire

can be magnetised/demagnetised at the flick of a switch

e.g. MRI scanners, speakers/earphones, Mag-Lev trains

Charges

- there are positive & negative charges
- measured in coulombs
- Like charges repel
- unlike charges attract

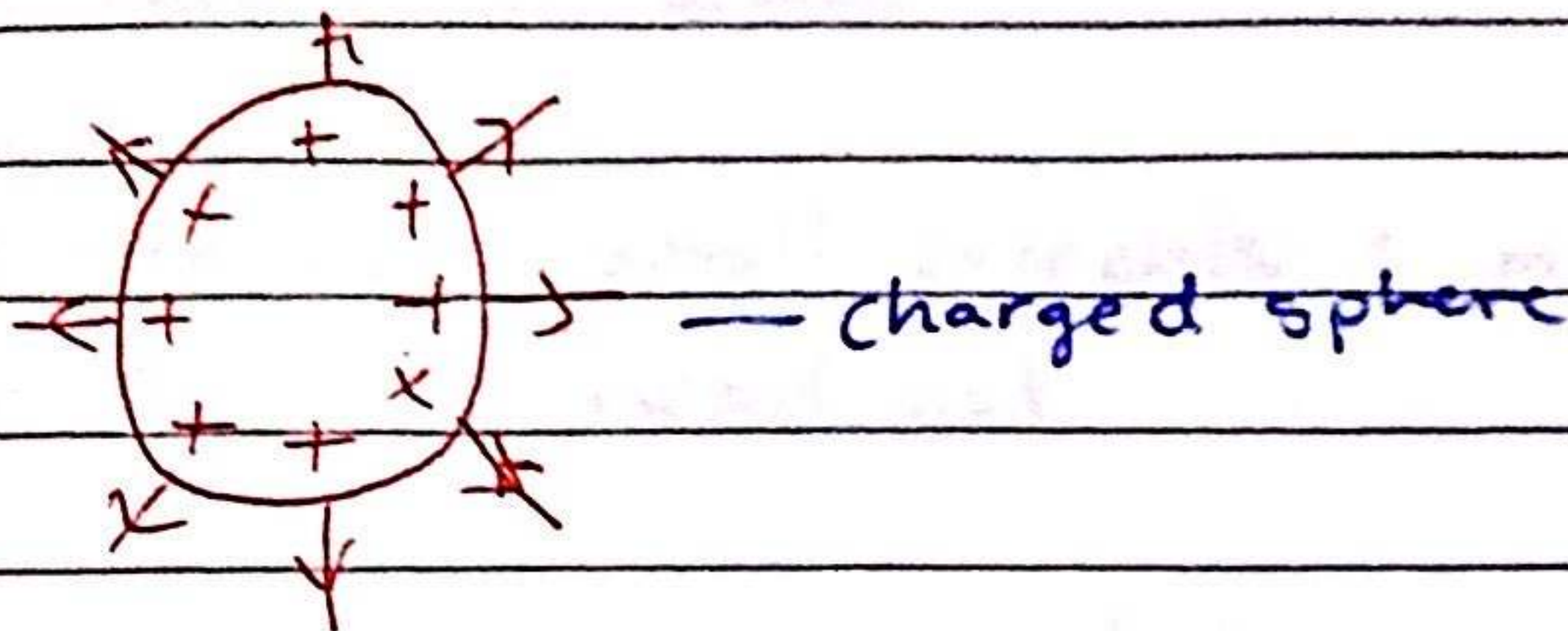
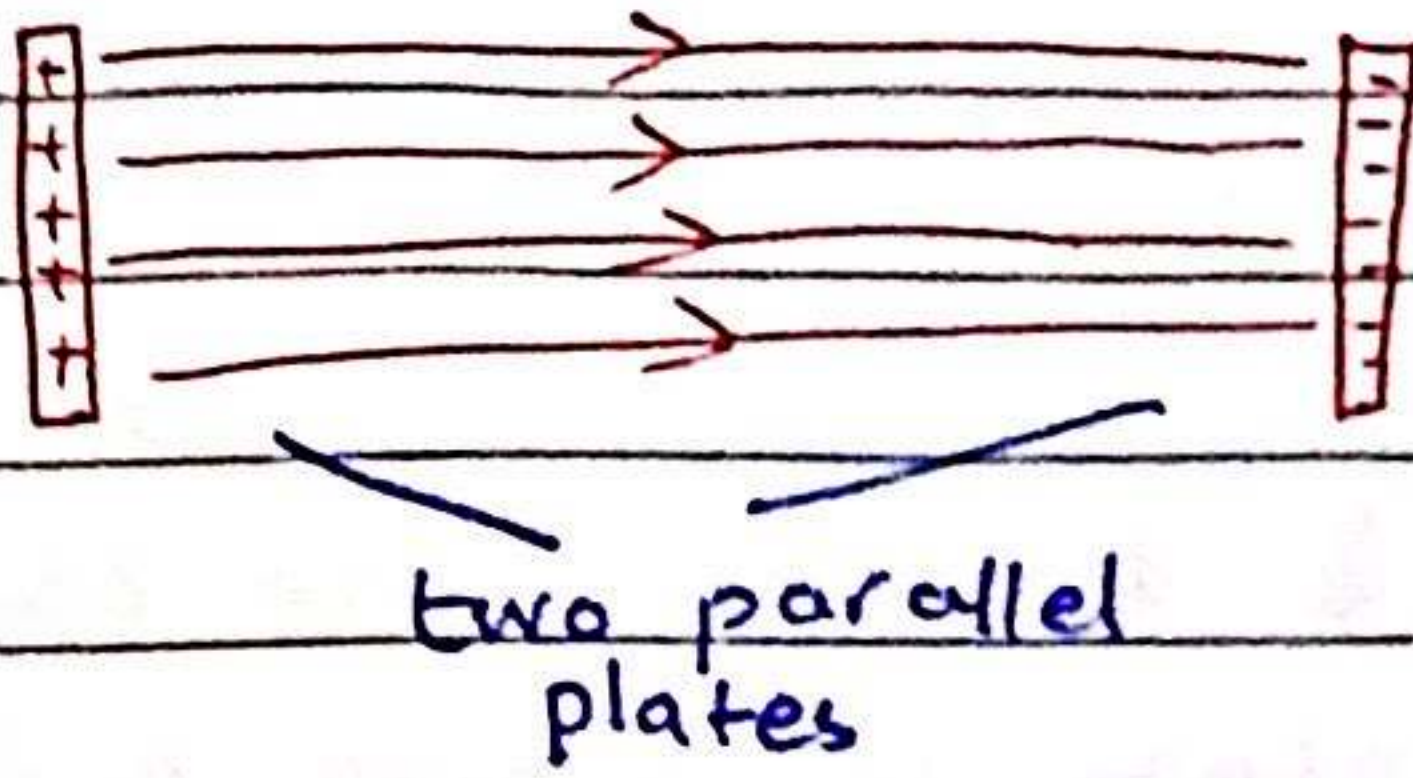
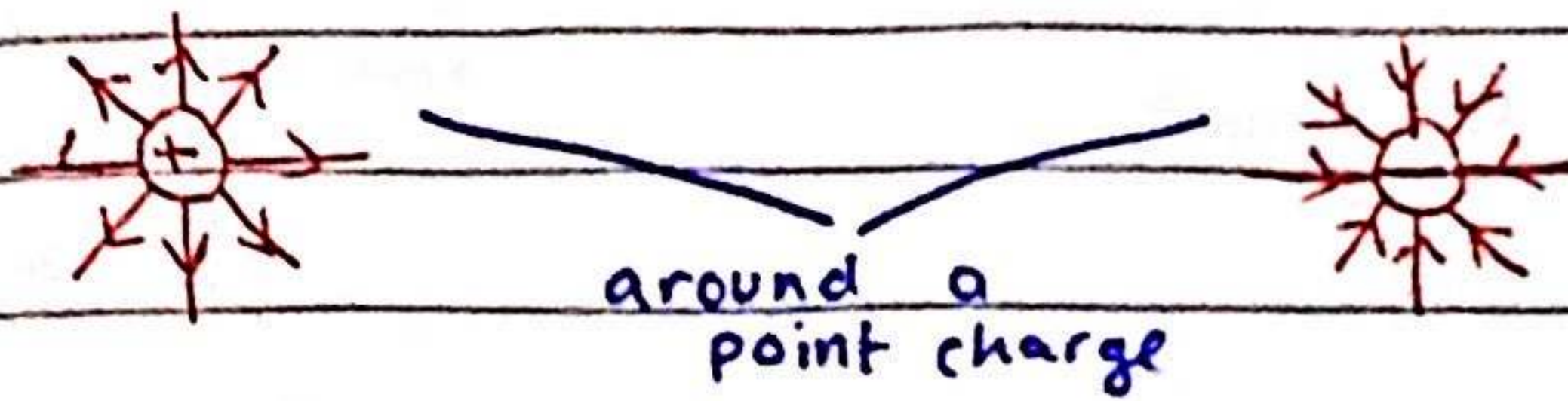
* an electric field is a region where electric charge experiences force

Demonstrating electrostatic charges

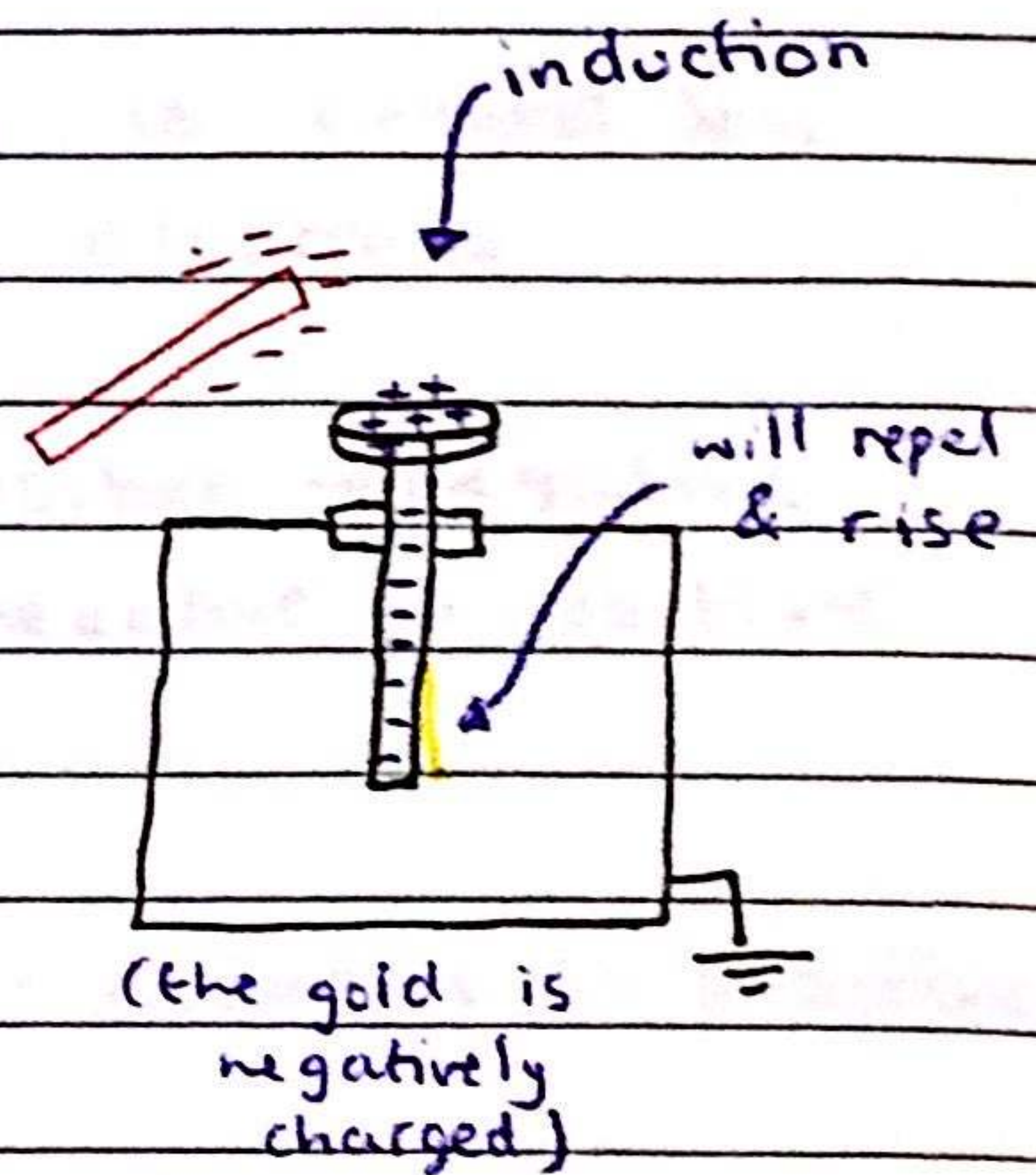
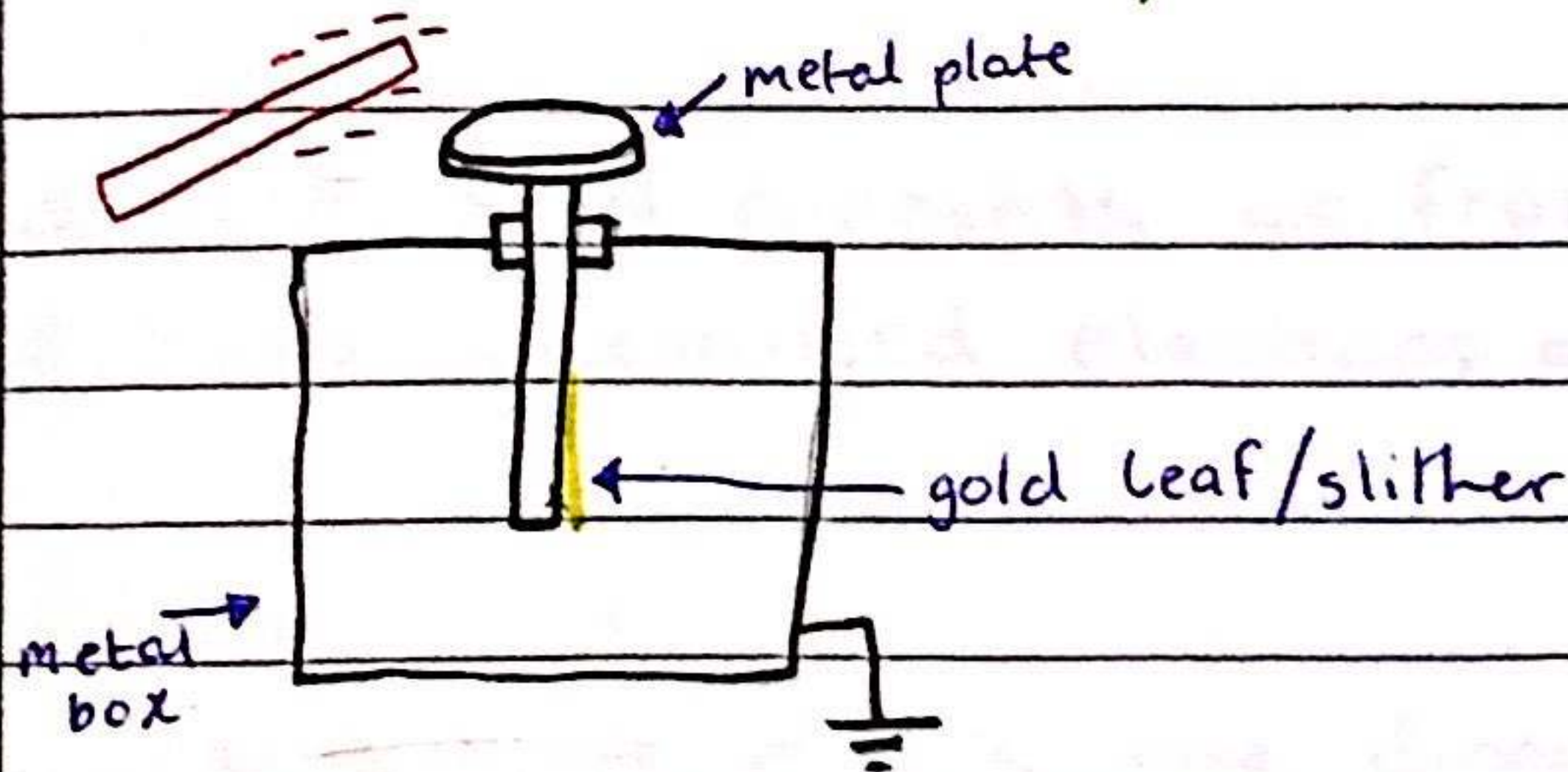
- by friction
 - ↳ rubbing a cloth on a glass rod
- by contact
 - ↳ a positively charged glass rod comes in contact with a neutral glass rod, some of the negative charge will go from the neutral to the positive one, thus both will be left slightly positive
- by polarization
 - ↳ a positively charged rod comes close to a metal conductive rod, the electrons in the metal conductive rod will be drawn towards the positive rod, now one side has a lot of electrons while the other has few, so we slice the rod, and you're left with one positive and one negative rod
- by grounding
 - ↳ negatively charged rod close to a neutral one, the neutral one becomes polarized with the positive side towards the negative rod. if we ground the rod then the repelled negative charges have a place to go: Earth.

Electric Field :-

- a charged object creates an electric field
- this can be shown by electric field lines



Gold-Leaf electroscope



- if the leaf falls then material is a good conductor, if it then it is a good insulator.

* Typically metals are very good conductors & non-metals are good insulators

Current

- the amount of charge passing a point in a circuit every second; charge per second

$$* I = \frac{Q}{t}$$

$\uparrow$ current $\uparrow$ time

(unit of current is 'A'/amperes)

ammeters

- can be both: digital & analogue

↳ analogue

- ranges are 0.1 - 1.0 A & 1.0 - 5.0 A
- with a needle or scale

↳ digital

- can measure very small currents: mA or μ A
- with an electronic read out

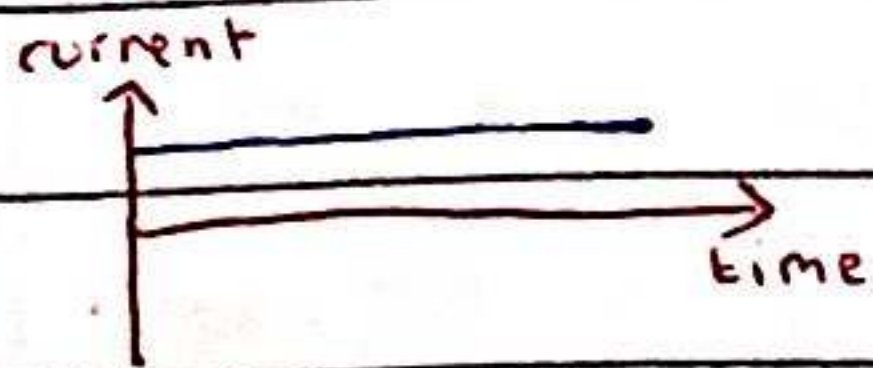
* electrical conduction in metals is caused by a flow of free (delocalised) electrons

* conventional currents go from positive $\rightarrow$ negative

* free delocalised electrons go negative $\rightarrow$ positive

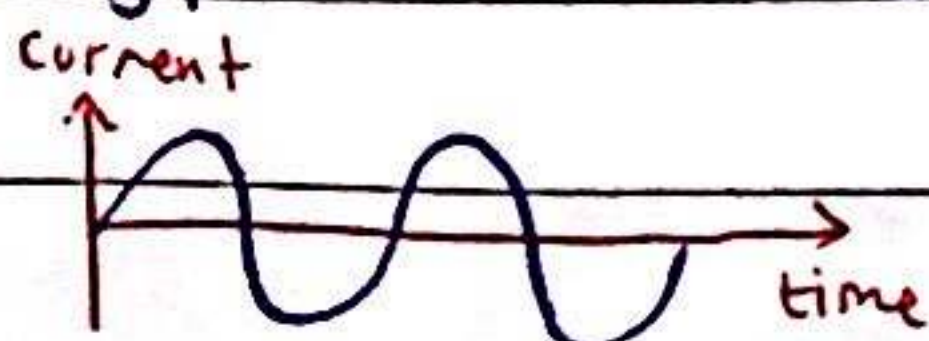
Direct current :-

- electrons flow in only one direction: negative to positive



Alternating Current :-

- electrons change direction regularly
- typical frequency in mains elec. is 50 Hz



Electromotive force (e.m.f) :-

- the electrical work done by a source in moving a unit charge around a complete circuit
- the voltage supplied by a power supply
- measured in volts (V)

$$* \underset{\substack{\uparrow \\ \text{e.m.f}}}{E} = \frac{W \leftarrow \text{Energy from the source (J)}}{Q \leftarrow \text{charge}}$$

Potential difference (p.d) :-

- work done by a unit charge passing through a component
- measured in (V) volts

$$* \underset{\substack{\uparrow \\ \text{p.d}}}{V} = \frac{W}{Q}$$

voltmeters

- can be both : analogue & digital

→ analogue

- ranges are 0.1 - 1.0 A and 0 - 5.0 A
- with needle or scale

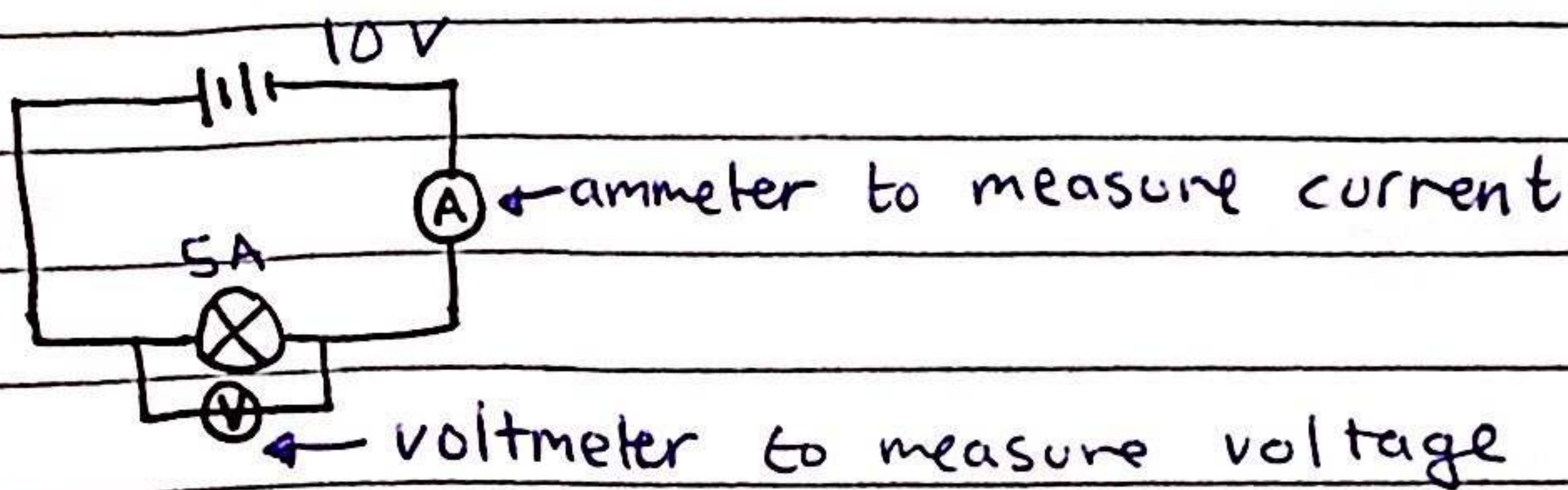
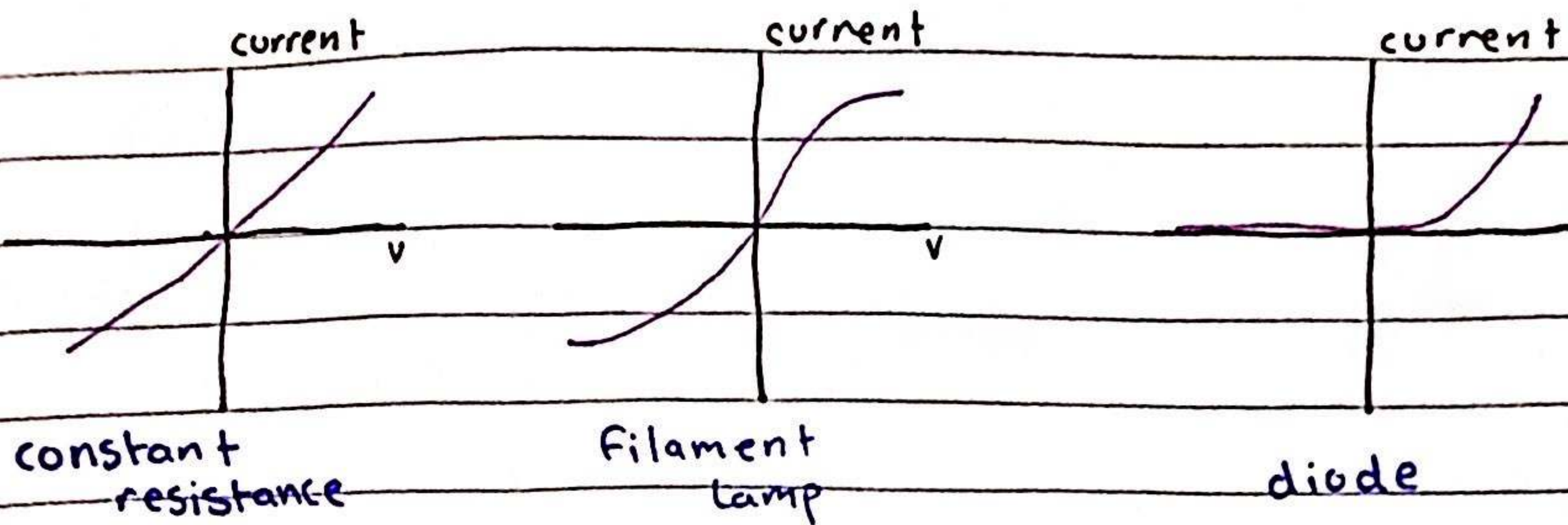
→ digital

- can measure very small potential difference: mV or μV
- easy & accurate
- with electronic read out

Resistance

- resistance is the opposition to current
- unit of resistance is 'ohm's' $\rightarrow \Omega$

$$* \underset{\substack{\uparrow \\ \Omega}}{R} = \frac{V \leftarrow \text{volts}}{I \leftarrow \text{current}}$$



- ① $R = V/I$
- ② $R = 10/5$
- ③ $R = 2 \Omega$

affect of ... on resistance :-

Length

- $R \propto L$
- Longer the wire, more the resistance

cross-sectional area

- $R \propto 1/A$
- more the cross-sectional area, less the resistance

* electric circuits transfer energy from a source to a component/device

$$\text{power} = P = IV$$

power
current
voltage

$$\text{Energy} = E = Pt$$

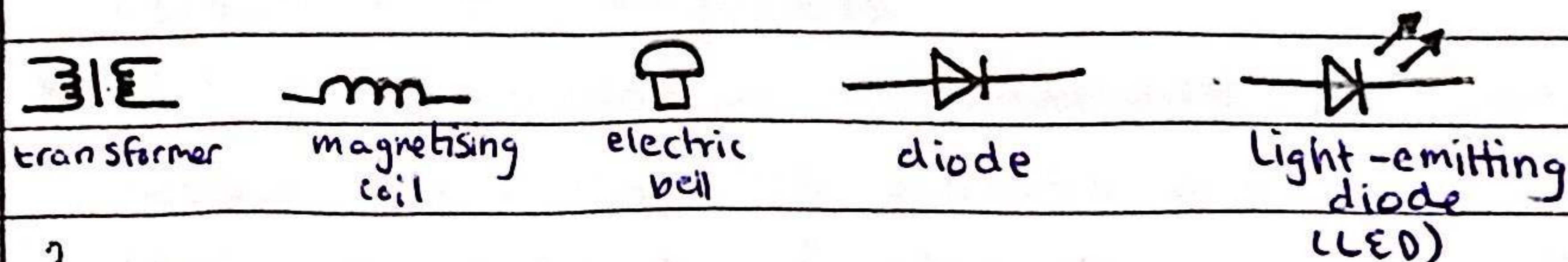
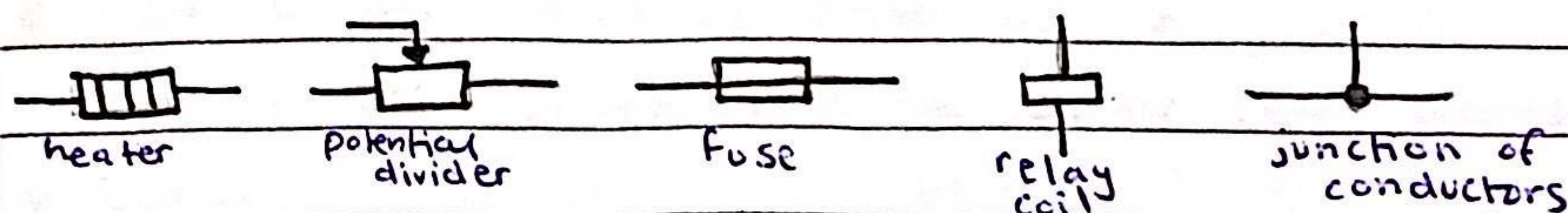
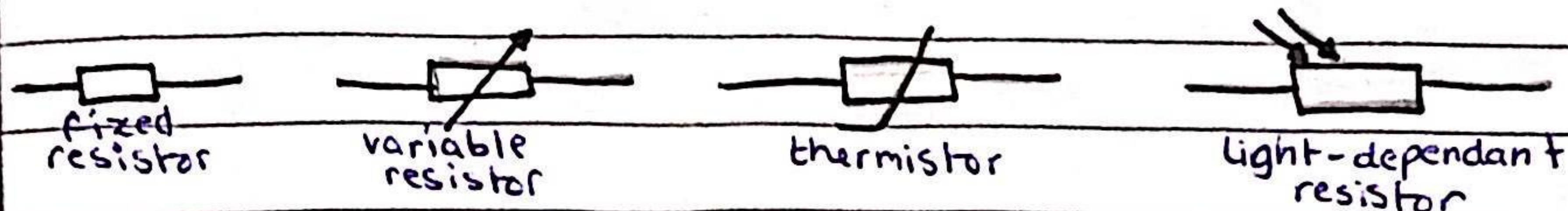
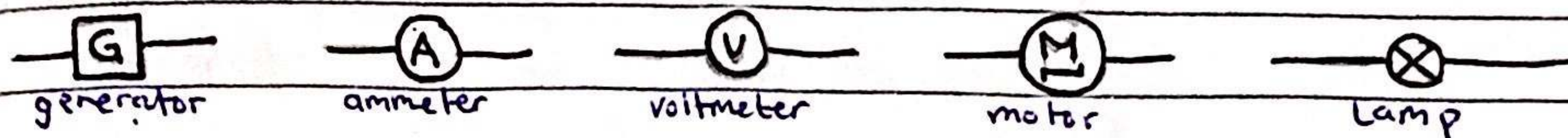
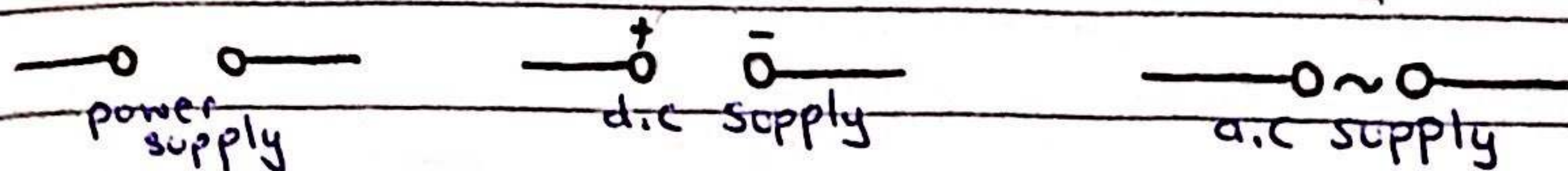
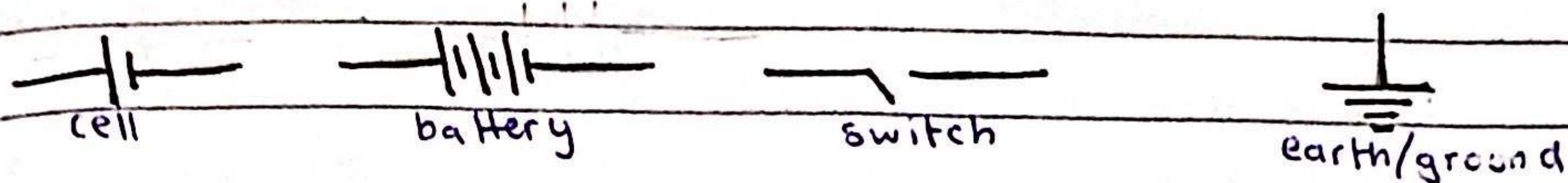
energy
power
time

$$1 \text{ kW} = 1000 \text{ W}$$

$$1 \text{ kWh} = 1000 \text{ Wh}$$

$$\rightarrow \text{cost} = \underline{\underline{\text{kW} \times t(\text{hrs})}}$$

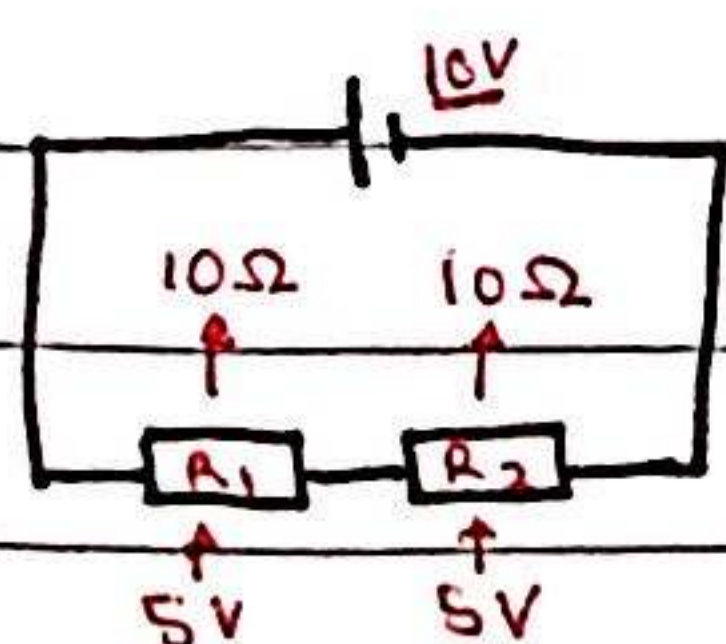
Circuits



* the current is the same at every point of a circuit. in a series connection!

Series connection

- total resistance of the circuit increases
- same current flows through all resistors
- the e.m.f. of the cell is divided across the resistors



$$* ① R_{\text{total}} = 10 + 10 = 20 \Omega$$

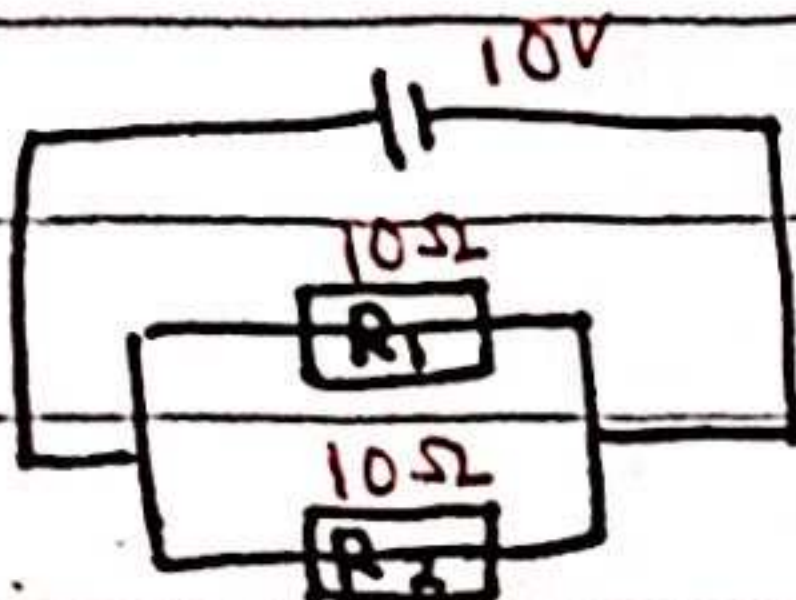
$$② I = VR = 10/20 = 0.5 \text{ A}$$

$$③ V_1 = IR_1 = 0.5 \times 10 = 5 \text{ V}$$

$$④ V_2 = IR_2 = 0.5 \times 10 = 5 \text{ V} \quad \text{--- } 10 \text{ V}$$

Parallel Connection

- total resistance of the circuit decreases
- current is divided between the resistors
- each resistor in the circuit gets the full voltage
- $\frac{1}{R_{\text{total}}} = \frac{1}{R_1} + \frac{1}{R_2} \dots$ OR $R_{\text{total}} = \frac{R_1 \times R_2}{R_1 + R_2}$ (2 resistors)



$$\textcircled{1} \frac{10 \times 10}{10 + 10} = 5 \text{ total}$$

$$R_{\text{total}} = 5 \Omega$$

$$\textcircled{2} I_{\text{total}} = V / R_{\text{total}}$$

$$I_{\text{total}} = 10 / 5 = 2 \text{ A}$$

$$\textcircled{3} I_1 = V / R_1 = 10 / 10 = 1 \text{ A} \quad I_2 = V / R_2 = 10 / 10 = 1 \text{ A} \quad \text{---} 2 \text{ A}$$

* Sum of the currents entering a junction ^(in a parallel circuit) =

Sum of the currents that leave the junction

* Total p.d. ^(in a series) = Sum of the individual p.d.s across each component

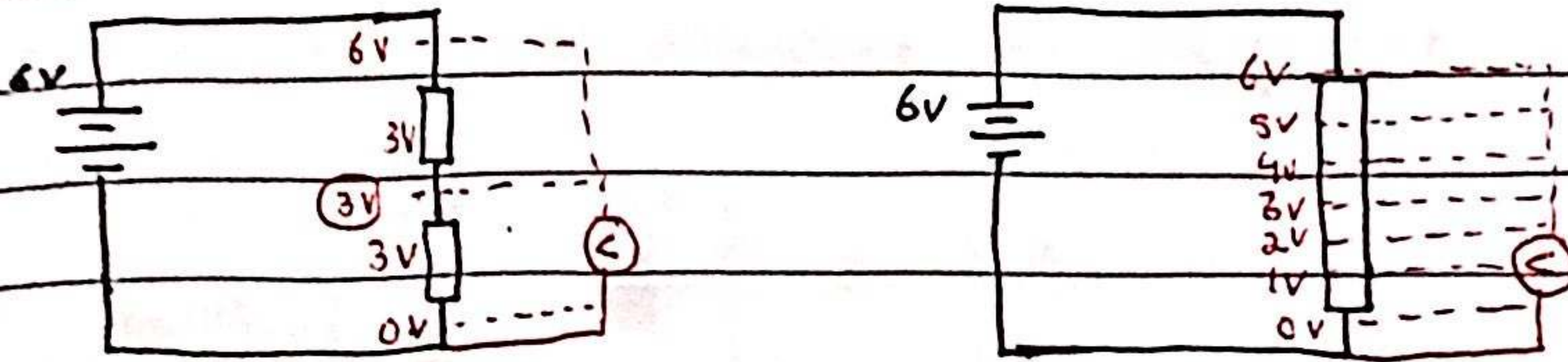
* The p.d. across an arrangement of parallel resistances = the p.d. across one branch in the arrangement of the parallel resistors

→ Two lamps in a parallel circuit enjoy the full voltage of the battery thus shining brighter than those in a series circuit.

* The p.d. across an electrical conductor increases as the resistance increases for a constant current

Variable Potential Divider (circuit):-

- a circuit used for selecting different output voltages



equation

$$* \frac{R_1}{R_2} = \frac{V_1}{V_2}$$

$$* V_{out} = \left(\frac{R_1}{R_1 + R_2} \right) \times V_{in}$$

Common hazards :-

- damaged insulation

if someone touches an exposed piece of wire, they could be subjected to lethal shock

- overheating of cables

passing too much current through a small wire can lead to overheating which could cause a fire or melt the insulations

- damp conditions

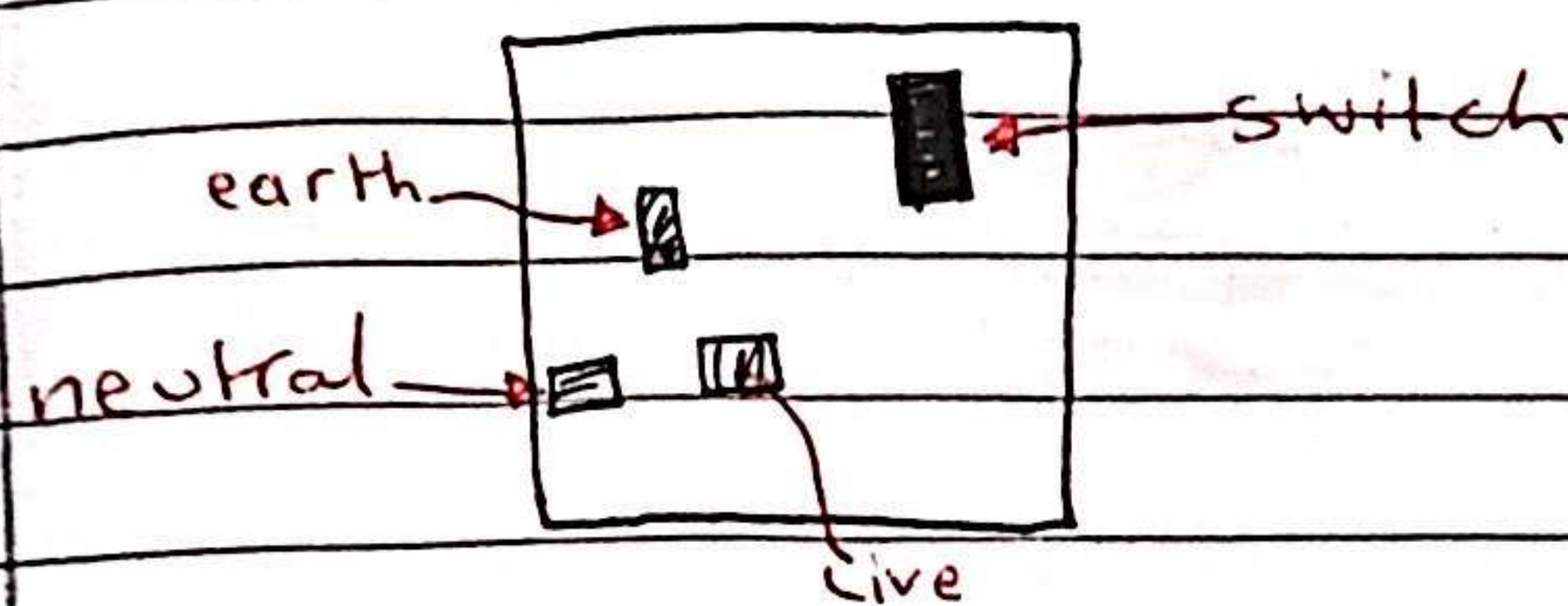
if moisture comes in contact with live wires, the moisture could conduct electricity either causing a short circuit, which could cause a fire or pose an electrocution risk.

- excess current from overloading of plugs, extension leads & sockets

if plugs, leads and sockets become overloaded due to plugging in too many components, the heat created can cause fires.

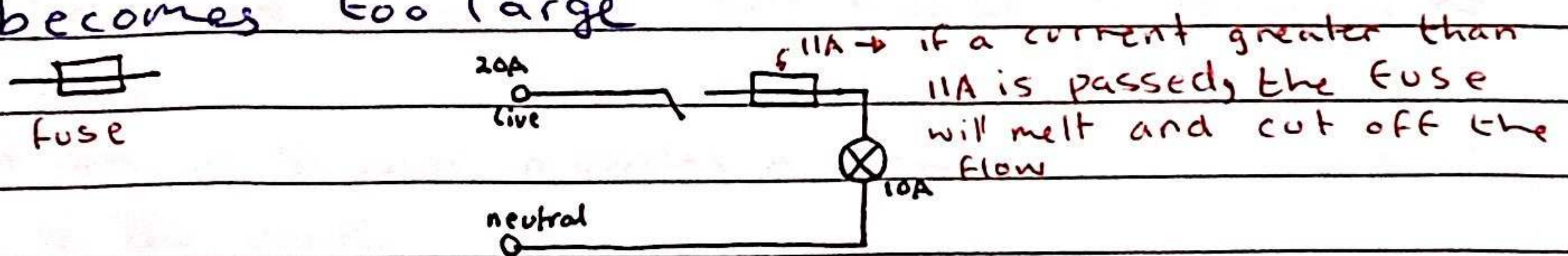
Mains electricity:-

- the electricity generated by power stations
- is an a.c supply
- 220-240V
- doesn't have positive or negative



Fuse:-

- a safety device designed to cut off the flow of electricity to an appliance if the current becomes too large



Trip switch:-

- when the current is too high the switch automatically flicks to the 'off' position.

what fuse to use ::

→ suppose an appliance uses 3.1 A

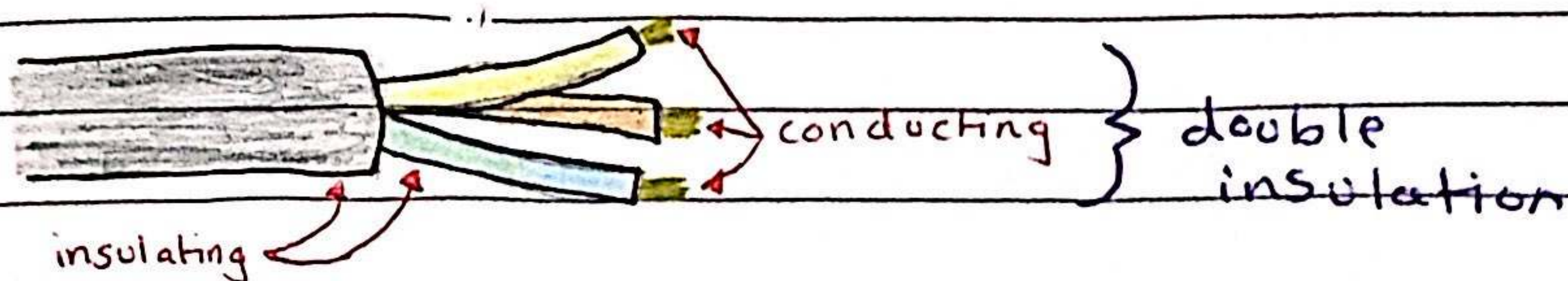
* 3 A would be too small

* 13 A would be too large

* 5 A would be an appropriate choice

Insulation & double insulation:-

- the conducting part of a wire is usually made of copper, if a person comes into contact with it, there is a risk of electrocution
- for this reason, wires are covered in an insulating material



Earthing:-

- many electrical appliances have metal cases
- if a live wire comes into contact with the case, the case would become electrified
- the earth wire is an additional safety wire

→ the earth wire provides a low resistance path to the earth

→ it causes a surge of current in the earth wire and hence also in the live wire

→ the high current goes through the fuse and causes it to melt & break which cuts off the supply of electricity.